DCR-RAG: Draft-Conditioned Ranking with Conservative Passage Replacement for Retrieval-Augmented Generation

Anonymous Author(s)

Abstract

Web answer engines use a language model, called the reader, to generate answers from a small set of retrieved passages. Selecting these passages is challenging because question-only ranking may miss indirectly connected evidence, while replacing passages to improve complementarity can discard useful support. We present DCR-RAG, a framework for Draft-Conditioned Ranking with Conservative Passage Replacement. In the ranking stage, a small language model generates a short answer draft from the candidate pool. A trainable cross-encoder uses this draft together with the question to score passages and select the initial pair. The scorer learns without gold answers from automatically constructed preferences that combine relevance with the decrease in draft log-likelihood after passage removal. In the replacement stage, an ensemble trained offline using gold answers estimates the reader-F1 gains of a small set of single-passage swaps. Candidates are ranked by predicted gain after penalizing disagreement among ensemble members, and the best is accepted only if its adjusted gain exceeds a threshold. Otherwise, the initial pair is retained. The reader then generates the final answer from the question and selected original passages, without receiving the draft. On five Wikipedia-based QA benchmarks, DCR-RAG achieves 54.46 macro F1 with two passages, outperforming the 8B SetR passage-set selector by 2.82 points. Holding the scorer fixed, replacement adds 0.92 F1 over directly selecting the two highest-scoring passages. At comparable replacement rates, the disagreement penalty reduces harmful swaps by about 40% relative to acceptance based on predicted gain alone. The code and data are available at https://anonymous.4open.science/r/DCR-RAG-75C3.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; **Question answering**; • **Computing methodologies** → **Natural language generation**.

Keywords

Retrieval-augmented generation, Web search, passage selection, draft conditioning, passage replacement

ACM Reference Format:

Anonymous Author(s). 2027. DCR-RAG: Draft-Conditioned Ranking with Conservative Passage Replacement for Retrieval-Augmented Generation. In *Proceedings of the ACM Web Conference 2027 (WWW '27), May 10–14, 2027, Dublin, Ireland*. ACM, New York, NY, USA, 12 pages.

1 Introduction

Answer engines and assistants built on Web search use retrieval-augmented generation (RAG) to ground their responses in retrieved content [13]. The search backend returns a pool of candidate passages, but the reader, the language model that generates the answer, usually receives only a small subset. A limited passage budget controls the reader's input cost, while longer contexts are not always used effectively [14]. Passage selection must therefore preserve the evidence needed for accurate answering within a compact context.

Two difficulties arise under a tight passage budget. First, evidence may be spread across entity-centric pages, and some passages may connect to the question through an entity that the question does not name. For example, answering which university the author of a novel attended requires identifying the author and then finding the author's educational background. The author's biography may provide the second fact without mentioning the novel. Under a two-passage budget, question-only scoring may undervalue this indirectly connected passage, allowing redundant descriptions of the novel to fill both slots. Second, changing the selected pair to improve complementarity can discard useful evidence. A more diverse pair does not necessarily produce a better answer, especially when the initial pair already provides sufficient support.

Prior work connects passage selection to reader behavior. GainRAG learns passage preferences from the gain each passage brings to generation [8]. SetR reasons about a question's information requirements and selects passages that satisfy them jointly [12], while R-CPS groups passages by the answers they induce in the reader [33]. These methods score passages against the question alone or build the passage set in a single pass, without testing whether a change improves on the context the reader would otherwise receive. Building on reader-oriented selection, we distinguish two decisions: how to rank passages with additional answer clues, and whether replacing a passage would improve on keeping the initial pair.

We propose DCR-RAG, a framework for Draft-Conditioned Ranking with Conservative Passage Replacement. For ranking, a small language model reads the candidate pool and produces a short answer draft. The draft can provide a tentative answer and entity or relation clues that help identify useful passages beyond those suggested by the question alone. A trainable cross-encoder then scores each passage jointly with the question and the draft. The scorer learns without gold answers from automatically constructed passage preferences that combine relevance with how much removing a passage lowers the drafting model's log-likelihood of the same draft. The two highest-scoring passages form the initial pair.

For replacement, DCR-RAG constructs a small set of candidate pairs, each formed by replacing one passage in the initial pair. An ensemble of five gain predictors estimates each candidate's change in reader F1 relative to that pair. The selector penalizes disagreement among their gain predictions and accepts the candidate with the highest adjusted gain only if that gain exceeds a threshold. Otherwise, it retains the initial pair. Gold answers are used offline to label replacement gains and tune the acceptance parameters. At inference, evaluating candidates requires no additional reader calls. The final reader receives only the question and the selected original passages. Although the draft is not passed directly to the reader, its errors can still affect selection and the final answer.

Across five Wikipedia-based QA benchmarks, DCR-RAG reaches 54.46 macro F1 under a two-passage budget, outperforming the 8B SetR passage-set selector by 2.82 points. Even without replacement, the scorer trained without gold answers outperforms the strongest baseline in macro F1. Holding this scorer fixed, replacement adds 0.92 F1 over directly selecting the two highest-scoring passages. At comparable replacement rates, the disagreement penalty reduces harmful replacements (those that lower answer F1) by about 40% compared with using predicted gain alone. The frozen selector also improves six additional readers without retuning.

Our main contributions are as follows:

- **Draft-conditioned ranking.** We condition passage scoring on both the question and a short draft generated from the candidate pool. The draft provides auxiliary ranking clues without occupying the final reader's passage budget.
- **Answer-free scorer supervision.** We construct passage preferences from relevance and the effect of passage removal on draft log-likelihood, enabling scorer training without gold answers.
- **Reference-relative replacement.** We learn reader-F1 gains for bounded changes to an initial passage pair and use disagreement-aware acceptance to decide whether to replace a passage or retain the pair.

2 Related Work

2.1 Reader-Oriented Retrieval and Reranking

DPR provides a foundation for passage retrieval in open-domain question answering [10], and RAG connects retrieval with downstream generation through a language-model objective [13]. Further studies address retriever–reader preference alignment, relevance-aware evidence use, and information refinement [5, 29, 35]. GainRAG estimates passage-level gain and trains a selector to align retrieval with the generator's preferences [8], and other approaches optimize reranking through language-model feedback and reinforcement learning [23, 31].

Large language models can also rerank directly or acquire ranking capabilities through training. RankGPT investigates listwise reranking [24], GenRT jointly models reranking and context truncation [34], and REARANK, Rank-R1, and ERank explore reasoning, supervised fine-tuning, and reinforcement learning for reranking [2, 37, 38]. DCR-RAG shares the goal of selecting evidence that benefits the reader but separates draft-conditioned ranking from the decision to revise an existing selection. Unlike these rerankers and GainRAG, which score passages against the question, our scorer conditions each score on the question, a pool-generated draft, and the passage, and a separate gain-prediction stage decides whether to change the resulting Top-2 pair or retain it.

2.2 Generated Context for Retrieval and Ranking

Iterative and adaptive RAG methods coordinate retrieval with intermediate reasoning or self-reflection [1, 26, 27]. Generated text can also enrich retrieval inputs. GAR expands queries with generated context [18], HyDE retrieves with the representation of a hypothetical document [6], query rewriting reformulates the question for retrieval [17], and MemoRAG guides retrieval with answer clues drafted from a global memory [21]. These approaches use generated content to construct or refine retrieval inputs.

Generated answers can also act on a fixed candidate list. RIDER moves passages that contain the reader's predicted answers to the top without training [19], and Speculative RAG verifies drafts written from distinct retrieved subsets with a larger model [30]. In both, the generated answer itself is matched or verified. DCR-RAG instead gives a draft generated from the candidate pool to a learned passage scorer, which uses its tentative answer and entity–relation clues to assess the original passages, so a passage can benefit from the draft without containing the answer. The draft is neither inserted into the evidence pool nor passed to the final reader, and therefore occupies no passage slot. Section 5.3 compares drafts with length-matched query expansion.

2.3 Passage Set Selection and Risk Control

Passage set selection evaluates whether passages jointly meet a question's information needs. SetR identifies these needs and selects passages that collectively satisfy them [12], while R-CPS groups passages by the answers they induce in the reader [33]. DCR-RAG instead considers bounded single-passage replacements of an independently scored Top-2 pair and predicts each alternative's reader-F1 gain over that pair, explicitly allowing the pair to be retained.

Risk-aware reranking assesses beneficial and harmful document effects through information gain [16], while counterfactual methods support evidence assessment and answer correction [15, 22]. In concurrent work, Return or Revise? predicts whether revising a draft answer beats returning it [20]. DCR-RAG asks this keep-or-change question about the reader's context and accepts a replacement only when its ensemble-predicted gain, penalized by predictor disagreement, exceeds a threshold. This conservative rule complements draft-conditioned ranking.

3 Methodology

3.1 Problem Formulation and Overview

Given a question q and a candidate pool $D = \{d_1, \ldots, d_N\}$ from a fixed retriever, DCR-RAG selects a set $S \subset D$ of original passages for a reader G. We study a budget of $|S| = 2$, where every slot matters, and extend it to four passages in Section 6.3. The objective is the quality of the answer $y = G(q, S)$ rather than the relevance of each passage in isolation.

Figure 1 and Algorithm 1 outline the pipeline. A frozen reference model G_{ref}, the drafting model, reads the pool and writes a draft $\hat{y}$. A scorer f_θ rates each passage given q and $\hat{y}$, and its two highest-scoring passages form the initial pair S_0 (Section 3.2). A selector proposes single-passage swaps to S_0 (Section 3.3), predicts their effect on the reader (Section 3.4), and replaces S_0 only when a swap is predicted to help reliably (Section 3.5). The reader then answers from q and S alone. The two stages rely on different supervision. The scorer learns from answer-free teacher preferences, and the gain predictor learns from reader outcomes measured against S_0. Both are trained offline, and inference uses no gold answer or reader output for a proposal.

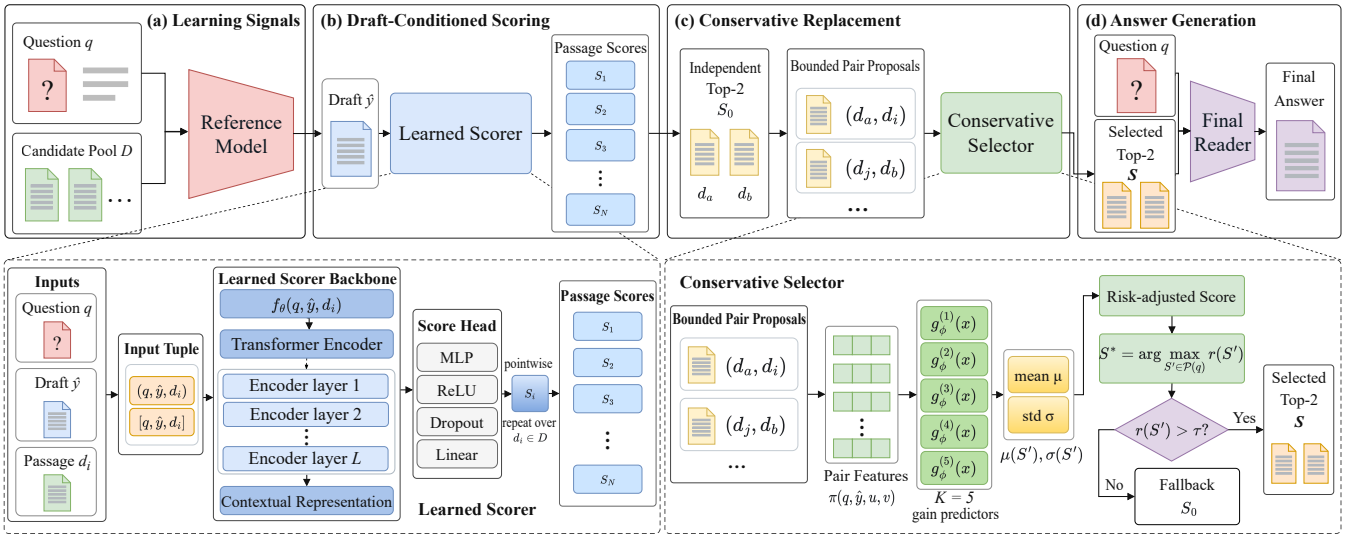


Figure 1: DCR-RAG: (a) offline learning signals and draft generation, (b) draft-conditioned scoring, (c) conservative replacement, (d) answer generation. Lower panels expand the scorer and the five-predictor selector. The reader never sees the draft.

3.2 Draft-Conditioned Passage Scoring

Draft generation. The reference model writes

$$\hat{y} = G_{\text{ref}}(q, D) \tag{1}$$

from the titles and texts of the candidates in BGE order, the order assigned by the frozen BGE reranker [32], within a 4,072-token input. We use Qwen2.5-3B-Instruct with greedy decoding and at most 24 output tokens. The resulting draft is a short tentative answer that often names the entities and relations behind it. Because G_{ref} has read the pool, the draft can carry information that the question lacks. In the example of Section 1, a draft that names the novel's author links the question to the passage about the author's education, even if it names the wrong university.

Scorer. The scorer f_θ, shown in the lower left of Figure 1, is a cross-encoder initialized from BGE-reranker-base. Its first segment holds the question and the draft, its second segment one candidate passage, and a score head maps the joint encoding to

$$s_i = f_\theta(q, \hat{y}, d_i), \qquad S_0 = \text{Top2}_{d_i \in D}\, s_i. \tag{2}$$

Ties follow BGE order. The complete draft appears verbatim in at most 1.10% of candidate passages on any dataset, so exact matching would rarely help. Joint encoding instead lets the scorer weigh partial draft–passage overlaps and learn how far to trust each clue.

Answer-free teacher. We train f_θ on preferences from a teacher that combines two signals. The frozen BGE score b_i captures topical relevance. The deletion response N_i captures how much the draft relies on d_i. It is the positive drop in the mean log-likelihood that the 3B reference assigns to the fixed draft when d_i is removed from the pool (Appendix A.2). The teacher utility is

$$u_T(q, d_i) = z_D(b_i) + 0.75 \min(N_i/c, 1), \tag{3}$$

where z_D is min–max normalization within the pool and c is the 90th percentile of positive responses on the dataset's scorer-training questions. The two signals fail in different ways. Relevance ignores what the draft depends on, and necessity inherits the draft's errors.

Algorithm 1 DCR-RAG inference

Require: Question q, pool D; fixed models and configuration

$\hat{y} \leftarrow G_{\text{ref}}(q, D)$
$s_i \leftarrow f_\theta(q, \hat{y}, d_i)$ for every $d_i \in D$
$S_0 \leftarrow \text{Top2}_{d_i \in D}\, s_i$; $S \leftarrow S_0$
$\mathcal{P}(q) \leftarrow$ deduplicated single-passage replacements
if $\mathcal{P}(q) \neq \varnothing$ **then**
  for each $S' = \{d_u, d_v\} \in \mathcal{P}(q)$ **do**
    $x_{S'} \leftarrow \pi(q, \hat{y}, u, v)$; compute $\mu(S'), \sigma(S')$
    $r(S') \leftarrow \mu(S') - \kappa\sigma(S')$
  end for
  $S^* \leftarrow \arg\max_{S' \in \mathcal{P}(q)} r(S')$
  if $r(S^*) > \tau$ **then**
    $S \leftarrow S^*$
  end if
end if
Order S by BGE rank; $y \leftarrow G(q, S)$
return S, y

Section 5.2 shows that their combination outperforms either one. The two passages with the highest utility form the teacher pair T_q, and every selected–unselected combination yields a preference,

$$\mathcal{B}_q = \{(i, j) : d_i \in T_q,\ d_j \in D \setminus T_q\}. \tag{4}$$

The scorer minimizes the logistic loss

$$\mathcal{L}_{\text{score}} = -\frac{1}{|\mathcal{B}_q|} \sum_{(i,j) \in \mathcal{B}_q} \log \text{sigmoid}(s_i - s_j). \tag{5}$$

The preference labels depend only on which two passages the teacher selects, not on its utility values, so the loss is a pairwise ranking surrogate for the two-slot decision at inference. It uses no gold answer.

3.3 Replacement Proposals and Pair Features

Let $S_0 = \{d_a, d_b\}$. For each position, DCR-RAG keeps the other initial passage and substitutes each of the four highest-scoring

passages outside S_0, which gives pairs of the form $\{d_a, d_i\}$ or $\{d_j, d_b\}$ with distinct passages. Removing duplicates yields the proposal set $\mathcal{P}(q)$ with at most eight pairs. Small pools yield fewer, and an empty set returns S_0. Keeping one initial passage bounds the search and preserves part of a context that the scorer already supports. Section 6.3 shows that wider search adds little.

Each proposal $S' = \{d_u, d_v\}$ is described by a feature vector

$$x_{S'} = \pi(q, \hat{y}, u, v). \tag{6}$$

The features cover three aspects of a pair. Pool-normalized scorer and BGE scores describe the preference for each passage. Question coverage, residual coverage, passage overlap, and length describe what the two passages contribute together. A draft-match flag records whether a passage contains the complete draft. For the question token set Q and passage token sets T_u, T_v,

$$\text{cover}(u, v) = \frac{|Q \cap (T_u \cup T_v)|}{\max(1, |Q|)}, \tag{7}$$

$$\text{residual}(v \mid u) = \frac{|(Q \setminus T_u) \cap T_v|}{\max(1, |Q \setminus T_u|)}. \tag{8}$$

Residual coverage counts the question tokens that d_v adds beyond d_u. A high value marks d_v as complementary to d_u.

3.4 Learning Reader Gains

For each gain-training question, the reader G answers from S_0 and from every proposal under the same budget and decoding settings. Scoring the answers against the gold aliases gives the gain

$$\Delta(S') = F_1\big(G(q, S')\big) - F_1\big(G(q, S_0)\big). \tag{9}$$

For a fixed question, ranking proposals by Δ or by absolute F1 gives the same order, since the F1 of S_0 is constant. Acceptance, however, compares each proposal with S_0, and zero gain is its reference point. We therefore regress Δ directly and call this formulation *Direct Gain*. The features describe only the proposal, and S_0 enters through the target. The predictor thus learns gains relative to the pairs that the learned scorer produces. Section 5.4 compares it with alternative formulations and tests transfer to other initial policies.

Each dataset provides 100 gain-training questions with about 700 labeled proposals, disjoint from the 100 scorer-training questions. We split the proposals into five folds grouped by question. For each fold k, a ridge regressor $g_\phi^{(k)}$ with $\alpha = 10$ is fitted on standardized features from the other four folds. We use ridge regression because labels are scarce and an MLP trained on them does not improve on it (Section 5.4). At inference, all $K = 5$ members score each proposal, and their mean and population standard deviation

$$\mu(S') = \frac{1}{K} \sum_{k=1}^{K} g_\phi^{(k)}(x_{S'}), \tag{10}$$

$$\sigma(S') = \sqrt{\frac{1}{K} \sum_{k=1}^{K} \big(g_\phi^{(k)}(x_{S'}) - \mu(S')\big)^2} \tag{11}$$

estimate the gain and the disagreement about it. Since each member omits a different fifth of the training questions, σ grows when a prediction depends on which questions the model has seen.

3.5 Conservative Acceptance and Answer Generation

The selector discounts each predicted gain by the disagreement,

$$r(S') = \mu(S') - \kappa\sigma(S'), \qquad \kappa \geq 0, \tag{12}$$

takes the best proposal, and accepts it only above a threshold:

$$S^* = \arg\max_{S' \in \mathcal{P}(q)} r(S'), \tag{13}$$

$$S = \begin{cases} S^*, & r(S^*) > \tau, \\ S_0, & \text{otherwise.} \end{cases} \tag{14}$$

The two possible errors have unequal costs. Rejecting a useful swap forgoes a gain, whereas accepting a harmful one damages a context that often already supports the answer. The rule therefore ranks swaps by the disagreement-penalized gain r rather than by μ alone, so a swap on which the members disagree needs a larger predicted gain to be accepted. The threshold τ sets the smallest predicted improvement worth a change. Development data select $\kappa = 1$ and $\tau = 0.04$, with gains on the $[-1, 1]$ scale, and both are frozen for testing.

Algorithm 1 summarizes inference. The reader receives the final pair in BGE order. For N candidates, scoring takes N cross-encoder passes, and selection evaluates five linear models on at most eight proposals instead of all $N(N-1)/2$ pairs. The reference model and the reader each run once per question.

4 Experimental Setup

4.1 Data, Candidate Pools, and Splits

We evaluate on three multi-hop datasets, HotpotQA [36], 2WikiMultiHopQA (2Wiki) [7], and MuSiQue [25], and two single-hop datasets, Natural Questions (NQ) [11] and TriviaQA [9]. All five are built on Wikipedia, and NQ questions are real queries issued to a Web search engine. The test set contains 1,000 questions per dataset and 500 for MuSiQue, 4,500 in total. For HotpotQA, 2Wiki, and MuSiQue, each pool is the candidate paragraph set released with the benchmark, with 2–10 passages on HotpotQA, 10 on 2Wiki, and 17–20 on MuSiQue. We neither add supporting passages nor use support annotations to build or order these pools. For NQ and TriviaQA, each pool holds the ten passages retrieved by E5 [28]. All methods share the passage IDs, texts, and BGE order. Test questions are drawn in hash order and exclude, by ID and normalized text, every question used for training, development, model selection, or confirmation. The MuSiQue subset comes from the 2,417-question MuSiQue-Ans v1.0 development split through validity checks, overlap removal, difficulty-oriented filtering, and hash-based sampling. Its 24 two-hop, 324 three-hop, and 152 four-hop questions put two-hop questions at 4.8%, against 51.8% in the source split [25], so it serves as a high-hop stress test rather than a sample of the benchmark distribution.

4.2 Models and Baselines

The reference model is Qwen2.5-3B-Instruct, the reader is Qwen2.5-7B-Instruct, and the scorer is initialized from BGE-reranker-base [32], an XLM-R cross-encoder [4]. All baselines select two passages from the same pools for the same reader:

Table 1: EM/F1 (%) on the main test set (n = 4,500) with shared pools, reader, and two-passage budget; macro averages weight datasets equally; best EM/F1 in bold. Last column: DCR-RAG minus method macro F1 [paired 95% CI].

Method	HotpotQA	2Wiki	MuSiQue	NQ	TriviaQA	Macro	ΔF_1 [95% CI]
BGE Top-2	62.40/72.71	23.90/38.83	9.80/19.87	23.50/38.34	53.10/64.78	34.54/46.90	+7.56 [6.57, 8.55]
BGE + MMR	61.80/72.28	23.20/37.76	10.00/19.98	23.80/38.66	53.50/65.24	34.46/46.78	+7.68 [6.70, 8.66]
BGE + same draft	64.60/75.16	26.00/42.01	12.60/23.21	24.50/39.41	53.30/65.80	36.20/49.11	+5.35 [4.42, 6.28]
GainRAG-controlled	55.80/65.93	21.60/35.90	12.60/20.92	27.60/42.69	53.80/65.36	34.28/46.16	+8.30 [7.29, 9.31]
REARANK-7B	62.70/73.58	33.00/50.16	13.00/24.23	28.10/42.85	55.40/66.85	38.44/51.53	+2.93 [2.04, 3.82]
ERank-4B	61.00/71.68	23.90/38.61	13.80/24.87	**30.50/45.80**	**56.10**/68.98	37.06/49.99	+4.47 [3.53, 5.41]
Rank-R1-3B	54.80/64.59	22.30/37.42	11.40/21.11	27.80/42.59	51.50/64.59	33.56/46.06	+8.40 [7.36, 9.44]
SetR-2 (controlled)	66.00/76.70	28.90/45.62	15.40/26.12	28.60/43.32	54.30/66.43	38.64/51.64	+2.82 [1.91, 3.74]
DCR-RAG	**66.20/76.82**	**33.80/52.06**	**16.80/30.08**	28.60/44.18	56.00/**69.16**	**40.28/54.46**	—

BGE Top-2; BGE with MMR [3]; BGE with the same draft appended to the query, without fine-tuning; GainRAG-controlled [8], fine-tuned per dataset without pseudo-passages because only original passages may reach the reader; the reasoning rerankers REARANK-7B [37], ERank-4B [2], and Rank-R1-3B [38]; and SetR [12] with its distilled Llama-3.1-8B-Instruct selector, which selects two passages from a 10,240-token context (Appendix A.1). We also compare with three full-pool pipelines, namely the 3B draft itself and the 7B reader answering directly from the full pool or revising the draft against it. Appendix B.1 adds a comparison with the released text-only DeBERTa critic of CoRM [15]. Per dataset, DCR-RAG uses disjoint sets of 100 scorer-training, 100 gain-training, and 100 development questions, and only the last two use gold answers. GainRAG-controlled trains on 100 questions with the same backbone, while SetR and the reasoning rerankers keep their distilled or released weights.

4.3 Training and Evaluation Protocols

Appendix A.2 gives the training settings. Unless stated otherwise, each controlled comparison changes only the factor under study, retraining affected components and reselecting acceptance parameters on the shared development splits. Input/output token limits are 4,072/24 for drafts, 2,048/32 for greedy reading, and 8,192/32 for the full-pool 7B pipelines. All runs use two RTX 4090 GPUs, and single-seed runs use seed 2094.

Metrics. We report exact match (EM) and token F1 of the complete normalized answer, maximized over gold aliases, and count generation failures as errors. Macro averages weight datasets equally, and differences are in percentage points. Intervals are 95% paired question-bootstrap intervals with 10,000 draws and seed 2092. Gain-prediction MAE averages $|\mu(S') - \Delta(S')|$ over all proposals, including rejected ones (Eq. 16). A replacement is harmful if it strictly lowers reader F1 relative to S_0. Acceptance and harm counts pool all 4,500 questions. Latency covers the request from the available pool to the final answer, excluding retrieval and model loading, on 250 fixed questions over three rounds after five warm-ups.

5 Experimental Results

5.1 Main Results and Held-out Confirmation

DCR-RAG achieves the best macro EM and F1 in Table 1, ahead of the two strongest baselines, SetR and REARANK-7B, by 2.82 and 2.93 F1. It has the best F1 on four of five datasets, and ERank-4B leads on NQ. The largest margins over SetR appear on 2Wiki and MuSiQue, where many questions chain relations through intermediate entities. It also exceeds GainRAG-controlled, the baseline closest to ours in supervision, by 8.30 F1. The gains are not confined to benchmark paragraph sets: on the E5-retrieved pools of NQ and TriviaQA, DCR-RAG improves on BGE Top-2 by 5.84 and 4.38 F1. Without the MuSiQue stress test, it still leads REARANK-7B and SetR by 2.2 and 2.5 macro F1. These baselines keep their released weights, so part of each margin may reflect DCR-RAG's target-domain supervision, but under the same supervision removing the draft lowers macro F1 from 54.46 to 52.57 (Table 3b).

Table 2: Held-out confirmation: EM/F1 (%) of the main checkpoints on 1,000 new questions per dataset; macro over HotpotQA and 2Wiki; best in bold.

Method	HotpotQA	2Wiki	Macro
BGE + same draft	64.30/74.90	25.80/41.96	45.05/58.43
SetR-2 (controlled)	65.50/75.76	28.70/45.76	47.10/60.76
Independent Top-2	64.20/74.82	28.90/45.10	46.55/59.96
DCR-RAG	**65.70/76.10**	**33.00/50.24**	**49.35/63.17**

Table 3: Input and teacher ablations: F1 (%), point estimates; best macro F1 in bold.

(a) Macro F1 by draft source and teacher

Configuration	Independent	Full
Pool draft + hybrid teacher	**53.54**	**54.46**
Question-only draft + hybrid teacher	51.96	52.84
Pool draft + BGE-only teacher	50.73	51.52
Pool draft + deletion-only teacher	51.82	52.69
Pool draft + hybrid, no backfill	53.50	54.42

(b) Per-dataset effect of the draft

Dataset	No draft	With draft	ΔF_1
HotpotQA	74.63	76.82	+2.19
2Wiki	49.44	52.06	+2.62
MuSiQue	28.64	30.08	+1.44
NQ	43.02	44.18	+1.16
TriviaQA	67.12	69.16	+2.04
Macro	52.57	54.46	+1.89

On 1,000 new questions each from HotpotQA and 2Wiki, the main checkpoints and frozen thresholds keep a 2.41-point macro-F1 lead over SetR [1.36, 3.47] (Table 2). This lead and the selector's 3.21-point gain over independent Top-2 from the same scorer both come mostly from 2Wiki, where the selector adds 5.14 F1.



Figure 2: Scorer × selection factorial with final checkpoints: independent Top-2 (Indep.) versus the full selector (Full). Dataset F1 at seed 2094; macro labels give three-seed means ± sample SD.

Table 4: Drafts versus exactly length-matched query expansion with the full selector, retrained with necessity-only labels; three-seed averages and paired 95% CIs.

Dataset	n	Draft	Expansion	ΔF_1	95% CI
HotpotQA	968	75.34	72.73	+2.61	[1.24, 3.98]
2Wiki	971	47.18	45.30	+1.88	[0.48, 3.29]
MuSiQue	473	24.45	22.93	+1.52	[0.08, 2.96]
NQ	958	41.47	40.41	+1.06	[0.06, 2.08]
TriviaQA	962	66.32	65.58	+0.74	[0.05, 1.44]
Macro	–	50.95	49.39	+1.56	[0.98, 2.14]

5.2 Scorer Learning and Teacher Supervision

Figure 2 crosses two scorers, BGE with the same draft and our learned scorer, with two selection routes, independent Top-2 and the full selector. Questions, pools, drafts, reader, and budget are shared, and the gain predictors of the main configuration serve both scorers without refitting. Because both scorers see the same draft, scorer learning measures what training adds beyond access to the draft. At seed 2094 it adds 4.43 F1 [3.51, 5.35] under independent selection, so the learned scorer alone, whose supervision uses no gold answers, reaches 53.54 macro F1 and already exceeds every baseline in Table 1. The full selector adds 0.92 [0.39, 1.45] on top of it, most on 2Wiki (2.10 F1, against 0.45–0.75 elsewhere). An interaction of only 0.03 [−0.48, 0.54] indicates that the two components contribute additively.

Table 3a varies the teacher with pool drafts fixed. The hybrid teacher exceeds the BGE-only and deletion-only teachers by 2.94 and 1.77 full-system F1, so relevance and draft necessity supply complementary supervision. In a separate control with a 7B reference and eight-passage groups, the answer-free necessity teacher performs on par with a teacher built on gold-answer likelihoods, and a richer structured teacher brings no measurable gain (Appendix B.2).

5.3 What Information Does the Draft Add?

Retraining the scorer without the draft costs 1.89 macro F1 [1.22, 2.57] and lowers F1 on every dataset (Table 3b). A draft generated from the question alone recovers little of this gain (panel a). Most of the draft's value therefore comes from the retrieved pool, not from the drafting model's parametric knowledge. On the held-out HotpotQA and 2Wiki questions, matched retraining confirms a draft gain of 2.99 macro F1 [2.07, 3.91] (Appendix B.2).

A draft also lengthens the scorer input. To separate content from length, a controlled variant compares drafts with query expansions that the same 3B model generates from the same context, cut to exactly the same length and retrained under identical settings. Drafts outperform expansions by 1.56 F1 [0.98, 2.14] on the 4,332 retained questions, with positive intervals on every dataset (Table 4). The draft's content, not the added input length, drives the gain.

Table 5: Fixed-scorer policies on the same proposals: macro F1 (%) and Direct Gain minus policy [paired 95% CI]. Shared inputs, budget, and development tuning.

Policy	F1	Direct Gain minus policy
Independent Top-2	53.54	+0.92 [0.39, 1.45]
MMR-style	53.63	+0.83 [0.21, 1.45]
Coverage	53.75	+0.71 [0.10, 1.32]
Combined	53.87	+0.59 [0.02, 1.20]

5.4 Replacement Decisions and Reliability

Learned gains versus pair heuristics. With the learned scorer and proposals fixed, we replace the gain model with MMR-style, coverage, and combined pair scores tuned on the same development data (Appendix B.3). Direct Gain outperforms all three, with paired intervals above zero (Table 5). The heuristics use the same coverage and overlap signals, but only reader-gain supervision tells when a change helps.

Gain formulation. Table 6 compares gain models trained on identical questions, proposals, and reader outputs. Predicting absolute F1 for both pairs and differencing the predictions performs worst. Regressing on feature differences, adding the initial pair's features, or switching to an MLP narrows the gap. Direct Gain still attains the lowest MAE, the highest F1, and the lowest harm rate among accepted swaps at nearly identical acceptance rates. Its F1 advantage over each alternative has a paired interval above zero. Encoding S_0 explicitly does not help, likely because each proposal keeps an initial passage whose scores already enter $x_{S'}$, and 700 labels per dataset favor fewer features.

Low MAE alone does not make swaps safe. Within Direct Gain, MAE falls as initial F1 rises, yet the harm rate among accepted swaps grows from 2.58% to 13.06% (Table 10b). A strong initial pair leaves little room for improvement, and small prediction errors then more often turn a swap harmful. The frozen gain models still improve initial pairs from the three heuristic policies, but by only 0.25–0.43 F1 (panel a). These policies start at most 0.33 F1 higher, while MAE rises from 0.164 to 0.178–0.190, so the smaller gains mainly reflect a shift from the training distribution.

Disagreement-aware acceptance. The full rule, with $\kappa = 1$ and $\tau = 0.04$, accepts a swap on 31.71% of questions and harms 2.82% (Figure 3). Removing the penalty ($\kappa = 0$) or the threshold ($\tau = 0$) accepts more swaps and harms more questions. The full rule outperforms these variants by 0.28 [0.04, 0.52] and 0.54 [0.22, 0.86] F1 while changing fewer contexts.

A stricter rule harms fewer questions simply by accepting fewer swaps. To isolate the penalty, we compare with a mean-only rule

Table 6: Gain formulations under shared supervision. F1 and rates in %; I: independent Top-2; Δ_I: F1 minus I; H/A: harmful among accepted swaps. MAE covers all valid proposals. Last column: full rule minus configuration [paired 95% CI].

Configuration	MAE	F1	Δ_I	Accept (%)	H/A (%)	Full minus configuration [95% CI]
Independent Top-2	—	53.54	—	0.00	—	+0.92 [0.39, 1.45]
Shared absolute-F1 difference	0.188	53.78	+0.24	31.82	13.06	+0.68 [0.15, 1.21]
Difference-feature gain	0.180	54.02	+0.48	31.67	12.00	+0.44 [0.05, 0.83]
Baseline-aware (three blocks)	0.174	54.20	+0.66	31.78	10.91	+0.26 [0.02, 0.50]
Baseline-aware (two blocks)	0.177	54.10	+0.56	31.73	11.76	+0.36 [0.04, 0.68]
MLP gain control	0.173	54.19	+0.65	31.71	10.93	+0.27 [0.02, 0.56]
Direct Gain (full rule)	**0.164**	**54.46**	+0.92	31.71	**8.90**	—

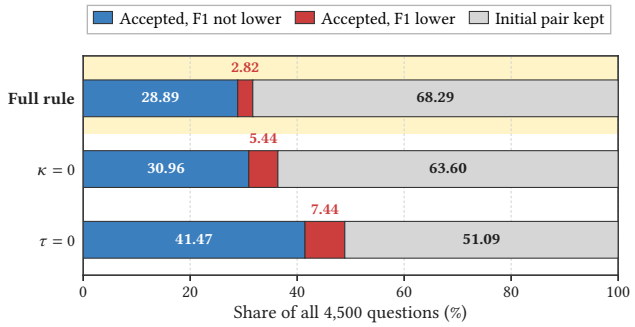


Figure 3: Replacement outcomes under the full rule and two ablations; accepted swaps that do not lower F1 include ties.

Table 7: Frozen-selector reader transfer, seed 2094. I/F: independent/full macro F1 (%); gain: paired F1 gain [95% CI]; H/A: harmful among the 1,427 accepted swaps (%).

Reader	I/F	Gain [95% CI]	H/A
Qwen2.5-1.5B	44.77/45.35	+0.58 [0.04, 1.12]	11.98
Qwen2.5-3B	48.08/48.60	+0.52 [0.08, 0.96]	11.42
Qwen2.5-7B	53.54/54.46	+0.92 [0.39, 1.45]	8.90
Qwen2.5-14B	52.45/53.14	+0.69 [0.12, 1.26]	9.74
Llama-3.1-8B	50.21/50.87	+0.66 [0.11, 1.21]	10.37
Mistral-7B-v0.3	45.10/45.60	+0.50 [0.01, 0.99]	12.75
InternLM3-8B	49.00/49.62	+0.62 [0.10, 1.14]	10.79

that sets $\kappa = 0$ and tunes its threshold on development data to match the full rule's acceptance rate. At nearly identical acceptance, the full rule makes 488 beneficial and 127 harmful replacements, and mean-only makes 444 and 210. The penalty thus removes about 40% of harmful replacements and raises macro F1 from 53.83 to 54.46, a gain of 0.63 [0.04, 1.22] that averages 0.59 ± 0.10 over three seeds (Table 14). The full rule's harmful swaps are also milder, lowering F1 by 10.9 points on average against 13.5 for mean-only. Mean-only keeps about a third of the full rule's 0.92-point gain over independent Top-2, and the full rule leads at every acceptance target from 20% to 50% (Table 11).

6 Robustness, Transfer, and Cost

6.1 Evidence Use and Draft Errors

Support recall is the fraction of annotated supporting passages that are selected, averaged over questions, and complete coverage requires all of them. On HotpotQA and 2Wiki, DCR-RAG raises complete coverage over BGE Top-2 from 82.30% to 89.20% and from 62.60% to 73.30%. Most of this gain comes from the scorer,

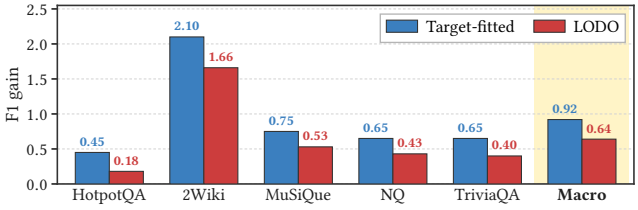


Figure 4: Selector gains over independent Top-2 with target-fitted and leave-one-dataset-out (LODO) gain models.

since independent Top-2 with the learned scorer already reaches 87.40% and 70.80%, and replacement adds the rest. On MuSiQue, two passages can fully cover only the 24 two-hop questions, and the gains reflect better partial support rather than complete reasoning chains. Support recall rises from 45.38% to 48.25% [1.33, 4.35], and on four-hop questions DCR-RAG shifts recall from the first hop toward the answer-bearing fourth, raising fourth-position recall by 15.13 points and lowering first-position recall by 7.89 (Table 12).

To test whether the reader uses the selected supports, we replace one selected passage with an equal-length passage while keeping the question, the other passage, and decoding fixed. On the 162 MuSiQue questions eligible for both interventions, replacing a supporting passage lowers F1 by 15.21 points and replacing a non-supporting one lowers it by 1.33. The paired difference is −13.88 [−17.12, −10.64]. Deletion instead of replacement, HotpotQA, and 2Wiki show the same pattern (Appendix C).

Draft errors. A wrong draft can mislead selection. Support exclusion, in which the final pair drops a gold supporting passage contained in the initial pair, occurs for 6.76% of questions with wrong natural drafts and 4.12% with correct ones. When we edit only the entity or only the relation of natural drafts, with pools, scorer, and reader fixed, strict-EM failures rise by 6.54 and 8.82 points (Appendix C). These failures arise in selection, since the reader never sees the draft.

6.2 Transfer across Readers and Datasets

We test whether the selector transfers to new readers (Table 7) and to unseen datasets (Figure 4).

Readers. We freeze the passages selected by the learned scorer, with and without the selector trained on Qwen2.5-7B labels, and give them to six other readers with matched prompts, decoding, and input budgets, without retraining or retuning. The selector improves all seven readers by 0.50–0.92 macro F1 with positive paired intervals, and harmful swaps are rarest for the training

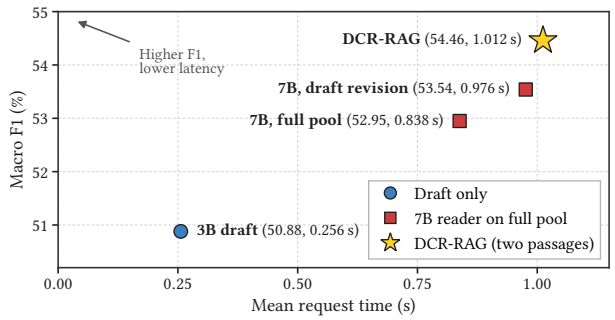
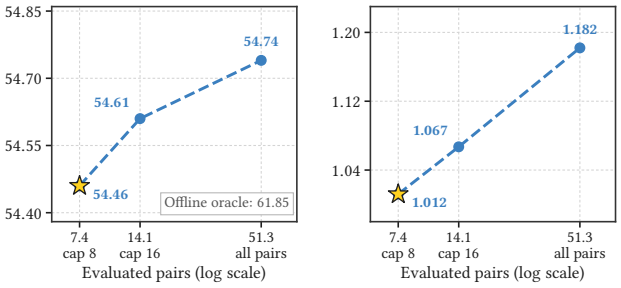


Figure 5: Full-pool pipelines and DCR-RAG, labeled with macro F1 and mean request time; quality on 4,500 and latency on 250 questions.



(a) Macro F1 (%) **(b) Mean request time (s)**

Figure 6: Search scope with 8- and 16-pair caps or all pairs: (a) macro F1; (b) mean request time on the timing subset. The offline oracle selects pairs by observed reader F1.

reader (Table 7). Draft conditioning, evaluated separately, adds 1.34–1.98 F1 across the same readers (Table 13a).

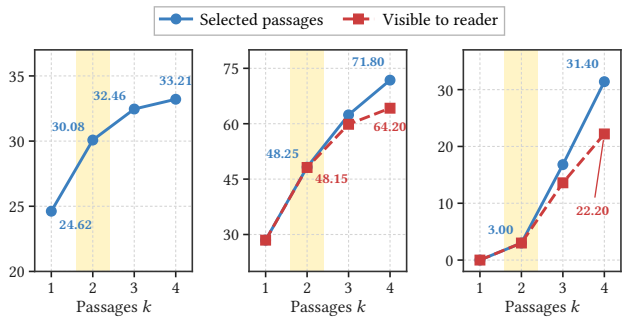
Datasets. In a leave-one-dataset-out (LODO) setting, we keep the target scorer but fit gain predictors, preprocessing, and acceptance parameters without any gain-training or development data from the target dataset. LODO improves every dataset and reaches 54.18 macro F1, 0.64 [0.28, 1.00] above independent selection and 0.28 below target fitting. The selection stage thus transfers across datasets, and in-domain labels add a further improvement.

6.3 Quality, Cost, and Passage Budgets

Latency. Because DCR-RAG also runs a 3B model over the pool, we time complete requests, which take 1.012 s on average and 1.148 s at the 95th percentile. Against pipelines in which the 7B reader processes the full pool (Figure 5), DCR-RAG gains 1.51 F1 over direct answering with about 21% more latency and 0.92 F1 over draft revision with 0.036 s more [0.018, 0.054]. The saving lies in the reader's input, two passages instead of the full pool, rather than in end-to-end latency, and matters most when the reader is large or billed per token.

Training cost. Supervision is a one-time offline cost of one reference-model pass per candidate passage on 100 questions and about 800 reader calls on another 100 per dataset.

Search scope. Doubling the proposal cap to sixteen pairs, or scoring all pairs, raises the average number of evaluated pairs from 7.4 to 14.1 and 51.3 but adds only 0.15 and 0.28 F1 while mean request time rises from 1.012 to 1.067 and 1.182 s (Figure 6). The eight-pair set already contains an exhaustive maximizer of the risk-adjusted score for 92.40% of questions. An offline oracle that selects pairs by observed reader F1 reaches 61.85 macro F1, leaving room for better gain prediction.



(a) F1 (%) **(b) Support recall (%)** **(c) Complete (%)**

Figure 7: MuSiQue passage budget: (a) F1, (b) support recall, (c) complete support coverage. Visible passages fit within the 2,048-token reader limit; the standard $k = 2$ is shaded.

Table 8: MuSiQue F1 (%) at matched budgets ($n = 500$, common reader). Point estimates.

Method	$k = 2$	$k = 4$
BGE Top-k	19.87	25.40
SetR-k	26.12	29.35
DCR-RAG	**30.08**	**33.21**

Passage budget. On MuSiQue, we vary the budget k without retraining or retuning. $k = 1$ keeps the top-scored passage, $k = 2$ the selected pair, and $k = 3, 4$ append the next highest-scoring passages, all in BGE order under the 2,048-token reader limit, so replacement itself is tested only at $k = 2$. With $k = 1$, the top-scored passage is already a gold support for 450 of the 500 questions, and additional slots serve the remaining hops. Moving from two to four passages adds 3.13 F1 for 0.396 s, while truncation increasingly hides support from the reader (Figure 7). DCR-RAG keeps its advantage at matched budgets, exceeding SetR by 3.96 F1 at $k = 2$ and by 3.86 at $k = 4$ (Table 8). With two passages, it already outperforms SetR with four.

7 Conclusion

DCR-RAG selects compact contexts for answering from Web content by separating how to rank the candidate pool from whether to revise the ranked pair. A draft written from the pool gives the scorer tentative answers and possible bridging entities without entering the reader's context, and a disagreement-penalized gain model decides whether a single-passage swap should replace the initial pair. With two passages, DCR-RAG reaches 54.46 macro F1 on five QA benchmarks, ahead of LLM rerankers and set selectors. The results suggest two design principles for compact RAG contexts: condition ranking on a hypothesis formed from the pool, and justify each change to the reader's context against the context already in hand. The main limitations are that draft errors can still mislead selection and that gain prediction requires gold-labeled swaps.

References

[1] Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International conference on learning representations*, Vol. 2024. 9112–9141.

[2] Yuzheng Cai, Yanzhao Zhang, Dingkun Long, Mingxin Li, Pengjun Xie, and Weiguo Zheng. 2026. ERank: Fusing Supervised Fine-Tuning and Reinforcement Learning for Effective and Efficient Text Reranking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 30121–30129.

[3] Jaime G Carbonell and Jade Goldstein. 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries.. In *SIGIR*, Vol. 98. 290941–291025.

[4] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th annual meeting of the association for computational linguistics*. 8440–8451.

[5] Guanting Dong, Yutao Zhu, Chenghao Zhang, Zechen Wang, Ji-Rong Wen, and Zhicheng Dou. 2025. Understand what LLM needs: Dual preference alignment for retrieval-augmented generation. In *Proceedings of the ACM on Web Conference 2025*. 4206–4225.

[6] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1762–1777.

[7] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*. 6609–6625.

[8] Yi Jiang, Sendong Zhao, Jianbo Li, Haochun Wang, and Bing Qin. 2025. Gain-RAG: Preference alignment in retrieval-augmented generation through gain signal synthesis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 10746–10757.

[9] Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1601–1611.

[10] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*. 6769–6781.

[11] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics* 7 (2019), 453–466.

[12] Dahyun Lee, Yongrae Jo, Haeju Park, and Moontae Lee. 2025. Shifting from ranking to set selection for retrieval augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 17606–17619.

[13] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.

[14] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics* 12 (2024), 157–173.

[15] Peiyang Liu, Qiang Yan, Ziqiang Cui, Di Liang, Xi Wang, and Wei Ye. 2026. Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generation. *arXiv preprint arXiv:2605.01302* (2026).

[16] Zhizhao Liu, Zhihua Wen, Zhiliang Tian, Zhen Huang, Miaorong Zhu, Zimian Wei, Yifu Gao, Liang Ding, and Dongsheng Li. 2026. Rethinking the Hidden Risk of Reranking: Achieving Risk-aware Reranking with Information Gain for RAG with LLMs. In *Proceedings of the ACM Web Conference 2026*. 1887–1898.

[17] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. 2023. Query rewriting in retrieval-augmented large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*. 5303–5315.

[18] Yuning Mao, Pengcheng He, Xiaodong Liu, Yelong Shen, Jianfeng Gao, Jiawei Han, and Weizhu Chen. 2021. Generation-augmented retrieval for open-domain question answering. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 4089–4100.

[19] Yuning Mao, Pengcheng He, Xiaodong Liu, Yelong Shen, Jianfeng Gao, Jiawei Han, and Weizhu Chen. 2021. Reader-guided passage reranking for open-domain question answering. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. 344–350.

[20] Nicholas Kashani Motlagh, Tim Anderson, Jeremy Gwinnup, and Grant Erdmann. 2026. Return or Revise? Learning When Revision Helps Retrieval-Augmented QA. *arXiv preprint arXiv:2609.30087* (2026).

[21] Hongjin Qian, Zheng Liu, Peitian Zhang, Kelong Mao, Defu Lian, Zhicheng Dou, and Tiejun Huang. 2025. Memorag: Boosting long context processing with global memory-enhanced retrieval augmentation. In *Proceedings of the ACM on Web Conference 2025*. 2366–2377.

[22] Huaiyu Qin, Chunyu Wei, Yueguo Chen, and Yunhai Wang. 2026. Counterfactual Reasoning for Retrieval-Augmented Generation. In *International Conference on Learning Representations*, Vol. 2026. 16683–16714.

[23] Jiashuo Sun, Xianrui Zhong, Sizhe Zhou, and Jiawei Han. 2026. Dynamicrag: Leveraging outputs of large language model as feedback for dynamic reranking in retrieval-augmented generation. *Advances in Neural Information Processing Systems* 38 (2026), 168661–168692.

[24] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. 2023. Is ChatGPT good at search? investigating large language models as re-ranking agents. In *Proceedings of the 2023 conference on empirical methods in natural language processing*. 14918–14937.

[25] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics* 10 (2022), 539–554.

[26] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*. 10014–10037.

[27] Liang Wang, Haonan Chen, Nan Yang, Xiaolong Huang, Zhicheng Dou, and Furu Wei. 2026. Chain-of-retrieval augmented generation. *Advances in Neural Information Processing Systems* 38 (2026), 59888–59915.

[28] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533* (2022).

[29] Yuhao Wang, Ruiyang Ren, Junyi Li, Wayne Xin Zhao, Jing Liu, and Ji-Rong Wen. 2024. Rear: A relevance-aware retrieval-augmented framework for open-domain question answering. In *Proceedings of the 2024 conference on empirical methods in natural language processing*. 5613–5626.

[30] Zilong Ryan Wang, Zifeng Wang, Long Le, Huaixiu Steven Zheng, Swaroop Mishra, Vincent Perot, Yuwei Zhang, Anush Mattapalli, Ankur Taly, Jingbo Shang, et al. 2025. Speculative rag: Enhancing retrieval augmented generation through drafting. In *International Conference on Learning Representations*, Vol. 2025. 18483–18505.

[31] Yuhang Wu, Xiangqing Shen, Fanfan Wang, Cangqi Zhou, Zhen Wu, Xinyu Dai, and Rui Xia. 2026. Optimizing RAG Rerankers with LLM Feedback via Reinforcement Learning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 30474–30490.

[32] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. 2024. C-pack: Packed resources for general chinese embeddings. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*. 641–649.

[33] Chunlei Xin, Shuheng Zhou, Xuanang Chen, Yaojie Lu, Huijia Zhu, Weiqiang Wang, Zhongyi Liu, Xianpei Han, and Le Sun. 2025. Aligning retrieval with reader needs: Reader-centered passage selection for open-domain question answering. In *Proceedings of the 31st International Conference on Computational Linguistics*. 1000–1012.

[34] Shicheng Xu, Liang Pang, Jun Xu, Huawei Shen, and Xueqi Cheng. 2024. List-aware reranking-truncation joint model for search and retrieval-augmented generation. In *Proceedings of the ACM Web Conference 2024*. 1330–1340.

[35] Shicheng Xu, Liang Pang, Mo Yu, Fandong Meng, Huawei Shen, Xueqi Cheng, and Jie Zhou. 2024. Unsupervised information refinement training of large language models for retrieval-augmented generation. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*. 133–145.

[36] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*. 2369–2380.

[37] Le Zhang, Bo Wang, Xipeng Qiu, Siva Reddy, and Aishwarya Agrawal. 2025. Rearank: Reasoning re-ranking agent via reinforcement learning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 2458–2471.

[38] Shengyao Zhuang, Xueguang Ma, Zheng Yao, Shuai Wang, Bevan Koopman, Jimmy Lin, and Guido Zuccon. 2026. Rank-r1: Enhancing reasoning in llm-based document rerankers via reinforcement learning. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 4419–4425.

A Implementation and Reproducibility

A.1 Models, Inputs, and Feature Processing

Execution settings. The draft model runs in vLLM with an 8,192-token context and a GPU memory fraction of 0.45 (token limits in Section 4.3); full-pool revision also receives the draft. GainRAG-controlled fine-tunes BGE-reranker-base on 100 questions per dataset (seed 2094) without pseudo-passages, and the common reader answers from its Top-2 passages. The anonymous repository, https://anonymous.4open.science/r/DCR-RAG-75C3, documents the SetR checkpoint, GainRAG checkpoints, MLP configuration, prompts, data IDs, seeds, and execution settings.

Controlled SetR configuration. SetR's distilled Llama-3.1-8B-Instruct selector reads a 10,240-token context with the candidates in BGE order and outputs two passages, which the common reader receives in BGE order. Decoding and output handling are in the repository.

Feature processing. Scorer and BGE scores are min–max normalized within each pool (constant pools map to zero), and pair indices refer to candidate positions fixed before reranking. Lexical features use Unicode NFD normalization, case folding, and Unicode `\w+` tokenization, keeping articles and stopwords; passage overlap is $|T_u \cap T_v|/\max(1, |T_u \cup T_v|)$. Draft match requires the complete normalized draft token sequence to occur contiguously in the passage title and text (empty drafts never match). It is nonzero for only 0.90, 0.65, 0.25, 0.55, and 1.10% of candidate passages on HotpotQA, 2Wiki, MuSiQue, NQ, and TriviaQA. Feature order is shared across datasets, and fitted normalizers and coefficients are frozen at inference.

A.2 Teacher Construction and Training

Main teacher. For Eq. 3, $\ell(\hat{y} \mid q, S)$ is the mean log probability that Qwen2.5-3B-Instruct assigns to the draft's answer tokens given q and passages S, excluding prompt tokens, padding, and an appended EOS token. The deletion response is

$$N_i = \max\{0, \ell(\hat{y} \mid q, D) - \ell(\hat{y} \mid q, D \setminus \{d_i\})\}. \tag{15}$$

Deletion keeps passage order and IDs and rebuilds the prompt under the 4,072-token limit, so previously truncated text can enter the context (backfill); the draft is not regenerated. BGE scores the question against the passage title and text. The quantile c is floored at 10^{-8} and set to one when no response is positive; empty drafts set $N_i = 0$. Ties follow BGE order and then ID; nonfinite utilities are excluded, with BGE Top-2 as fallback if fewer than two remain.

Scorer and gain training. The 100 scorer-training questions per dataset come from HotpotQA distractor validation, 2Wiki train, MuSiQue-Ans train, and the E5 retrieval files `train_1w.json` for NQ and TriviaQA. Checks by ID and normalized text exclude all evaluation questions. AdamW updates all scorer parameters for three epochs (learning rate 10^{-6}, linear schedule with 0.1 warm-up, weight decay 0.01, one question group per device, four accumulation steps, FP16, gradient checkpointing); the final epoch provides the checkpoint. The question–draft segment and the passage are pretruncated to 200 and 464 tokens and jointly encoded with a 512-token limit and `only_second` truncation. Each group has at least three candidates, and preferences are averaged within the group. Gain-training folds are $f(q) = \text{int}[\text{SHA256}(s \| \text{dataset}(q) \| \text{id}(q))] \bmod 5$ with a frozen salt s. Each ridge member fits its own standardizer and weights on four folds ($\alpha = 10$, with intercept; constant features keep unit scale). Stability runs at seeds 2094–2096 retrain the scorer and gain predictors with generators, reader, pools, splits, prompts, and scoring fixed.

B Evaluation and Controls

B.1 Evaluation and System Comparisons

Evaluation and uncertainty. Complete answers are scored against all gold aliases without post-hoc extraction; HotpotQA follows the official yes/no/noanswer rules, and the empty normalized NQ alias and generation failures remain in the denominators. Pointwise bootstrap intervals resample questions within each dataset with route pairing and then macro-average; seed-averaged intervals first average same-question effects across seeds; joint intervals resample questions and seeds with pairing. Gain-prediction MAE covers the main-test questions Q_d^+ of dataset d with nonempty proposal sets, including rejected proposals, with reader F1 in $[0, 1]$:

$$\text{MAE} = \frac{1}{5}\sum_{d=1}^{5} \frac{1}{|Q_d^+|} \sum_{q \in Q_d^+} \frac{\sum_{S' \in \mathcal{P}(q)} |\mu(S') - \Delta(S')|}{|\mathcal{P}(q)|}. \tag{16}$$

System comparisons. On the same 2,500 HotpotQA, 2Wiki, and MuSiQue questions and BGE display order, CoRM and DCR-RAG reach 66.45/76.82, 31.67/52.06, and 25.46/30.08 F1, with paired gain intervals $[8.05, 12.70]$, $[17.95, 22.81]$, and $[1.30, 7.95]$; the macro gain is 11.79, approximately $[10.22, 13.36]$. In the confirmation experiment of Table 2, DCR-RAG exceeds SetR by +0.34 $[-1.11, 1.79]$ F1 on HotpotQA and +4.48 $[2.95, 6.01]$ on 2Wiki; the full selector exceeds independent Top-2 with the learned scorer by 3.21 $[2.15, 4.28]$ macro F1 and BGE with the same draft by 4.74 $[3.69, 5.80]$.

B.2 Scorer Supervision and Matched Inputs

Truncation backfill. The no-backfill variant deletes passages only from the originally truncated context, so no new text can enter; teacher preferences, scorer, gain predictors, and acceptance parameters are refitted with all else fixed (Table 3a).

7B-reference teacher control. On HotpotQA and 2Wiki, an independent 7B reference with its own drafts supervises scorers under three teachers. Each group holds the eight passages that frozen BGE ranks highest in the question's original pool, without adding or removing supporting passages. Training questions, initialization, loss, candidate order, Top-2 output, and optimization and truncation settings are matched, and models are frozen before testing on the same held-out questions for all teachers and seeds. The reader, prompt, and decoding match the main experiments, so absolute F1, including that of the BGE anchor, differs from Table 1 only through these questions and their candidate groups. With ℓ the mean reference log-likelihood over nonpadding answer tokens, Scalar uses $U_i^{\text{scalar}} = \ell(a_q^* \mid q, \{d_i\}) - \ell(a_q^* \mid q, \varnothing)$ for a fixed gold alias a_q^*, and Necessity uses Eq. 15 with the non-oracle draft. Structured anchors on the normalized consensus of BGE Top-3/Top-2 and MMR Top-2; with the conflict-minus-anchor margin $m(S)$, it

Table 9: 7B-reference teacher control on BGE Top-8 groups and its own 1,000 held-out questions per dataset: F1 (%), mean ± sample SD over seeds 2094–2098.

Teacher / anchor	HotpotQA	2Wiki	Macro
BGE anchor	71.82	56.74	64.28
No-draft scorer	72.91	58.06	65.48
Scalar	74.09 ± 0.06	62.24 ± 0.04	68.17 ± 0.04
Necessity	74.13 ± 0.07	62.25 ± 0.06	68.19 ± 0.05
Structured	74.15 ± 0.05	62.20 ± 0.06	68.18 ± 0.05

defines $H_i = \max\{0, m(S \cup \{d_i\}) - m(S)\}$ for $d_i \notin S$, and a type-preserving minimal edit $s \to t$ gives $m_{t,s}(d) = \ell(t \mid q, d) - \ell(s \mid q, d)$ and $C_i = \max\{0, m_{t,s}(d_i^{\mathrm{cf}}) - m_{t,s}(d_i)\}$. Masks m_i^H, m_i^C encode consensus, source uniqueness, insertion, edit validity, and direction; invalid or unavailable signals and masks are zero, whereas valid zeros keep mask one. With the raw BGE score U_i^{sem}, pool min–max z_D (zero for constant pools), and training standardization z_{tr} (unit scale for constant features),

$$z_i = [z_D(U_i^{\mathrm{sem}}), z_{\mathrm{tr}}(N_i), -m_i^H z_{\mathrm{tr}}(H_i), \\ m_i^C z_{\mathrm{tr}}(C_i), m_i^H, m_i^C]. \tag{17}$$

A linear head fitted on training data yields the Top-2 labels; likelihoods are computed offline, ties follow BGE order and ID, and BGE Top-2 is the fallback when fewer than two finite scores remain. Necessity–Scalar, Structured–Scalar, and Structured–Necessity differ by +0.02 $[-0.07, 0.11]$, +0.01 $[-0.08, 0.10]$, and −0.01 $[-0.10, 0.08]$ F1 (Table 9), with Top-2 agreement of 93.1/92.7/94.4% and preference conflict of 5.8/6.2/4.9%. The answer-free Necessity teacher thus performs on par with one built on gold-answer likelihoods.

Matched drafts and query expansion. Both arms generate text greedily with Qwen2.5-3B from the same candidate context (128-token limit). The auxiliary text, i.e., the draft in one arm and the expansion in the other, is cut to a whole-word prefix so that both arms have equal token counts for the auxiliary text and for the complete query (question plus auxiliary text), capped at 64 and 200 tokens, without filler. Matching retains 4,332 test questions and 95/97 HotpotQA/2Wiki training questions, whose matched auxiliary texts average 13.20/11.58 tokens in both arms. Arms are retrained at seeds 2094–2096. Intervals average same-question outcomes across seeds before paired question resampling; the macro interval averages the five dataset gains within each replicate (Table 4). Draft/expansion support recall is 92.23/90.50% on HotpotQA and 88.79/87.01% on 2Wiki.

Matched-input confirmation. This confirmation retrains matched no-draft and full-draft arms, so its absolute F1 is not comparable with the main checkpoints of Table 2. Over three seeds, no-draft/full-draft F1 is 71.84/74.52 on HotpotQA and 38.71/42.02 on 2Wiki, gains of 2.68 $[1.43, 3.91]$ and 3.31 $[1.86, 4.74]$; macro F1 rises from 55.28 to 58.27, +2.99 $[2.07, 3.91]$, joint CI $[1.84, 4.12]$.

B.3 Selection and Decision Controls

Fixed-scorer heuristics. Let $p_i = z_D(s_i)$, $A = (p_u + p_v)/2$, J the Jaccard overlap of the two passages, $C = \mathrm{cover}(u, v)$, and $R = [\mathrm{residual}(v \mid u) + \mathrm{residual}(u \mid v)]/2$ (Section 3.3). The pair scores

Table 10: Selection diagnostics on 4,500 questions; F1 and rates in %. In (b), strata use F1 on the $[0, 1]$ scale, MAE is computed within each stratum, accepted swaps are given as n (%), and H/A is harmful among accepted.

(a) Frozen gain models applied to other initial policies

Policy	Initial F1	After F1	Gain	MAE	95% CI
Top-2	53.54	54.46	+0.92	0.164	[0.39, 1.45]
MMR-style	53.63	54.06	+0.43	0.178	[0.12, 0.74]
Coverage	53.75	54.10	+0.35	0.184	[0.07, 0.63]
Combined	53.87	54.12	+0.25	0.190	[0.02, 0.48]

(b) Direct Gain by initial-F1 stratum

Initial F1	n	MAE	Accepted	Harmful	H/A
[0, 0.25)	1,000	0.203	310 (31.00)	8	2.58
[0.25, 0.50)	900	0.179	280 (31.11)	16	5.71
[0.50, 0.75)	800	0.157	255 (31.88)	27	10.59
[0.75, 1.00]	1,800	0.137	582 (32.33)	76	13.06
Total	4,500	—	1,427 (31.71)	127	8.90

Table 11: Acceptance at development targets: main κ (full rule) or $\kappa = 0$ (mean-only), frozen per-dataset thresholds. Accepted: n (% of questions); H/A: harmful/accepted (%).

	Full rule			Mean-only		
Target	Accepted	H/A	F1	Accepted	H/A	F1
20%	906 (20.13)	6.7	54.24	898 (19.96)	9.8	53.80
25%	1,132 (25.16)	7.8	54.36	1,119 (24.87)	12.5	53.85
31.7%	1,427 (31.71)	8.9	54.46	1,431 (31.80)	14.7	53.83
40%	1,816 (40.36)	11.3	54.35	1,805 (40.11)	18.0	53.77
50%	2,241 (49.80)	15.4	54.18	2,262 (50.27)	22.0	53.64

are

$$H_{\mathrm{MMR}} = A - \lambda J, \qquad H_{\mathrm{Cov}} = A + \beta C, \tag{18}$$

$$H_{\mathrm{Comb}} = A + \beta C + \gamma R - \delta J. \tag{19}$$

Each heuristic maximizes H over $\{S_0\} \cup \mathcal{P}(q)$, so the MMR-style rule compares pairs jointly; ties keep S_0, then follow BGE order and ID. Coefficients are chosen from $\{0, 0.25, 0.5, 0.75, 1\}$ by development macro F1 (five settings each for MMR and coverage, 125 for the combined rule) and frozen; no reader-gain labels are used.

Reference-pair formulations. Difference-feature gain regresses $\Delta(S')$ on $x_{S'} - x_{S_0}$ with the five-fold ridge protocol; shared absolute-F1 difference fits one ridge predictor h on absolute reader F1 and predicts $h(x_{S'}) - h(x_{S_0})$. Baseline-aware gain uses

$$x_{S'}^{\mathrm{BA}} = [x_{S'}; x_{S_0}; x_{S'} - x_{S_0}], \tag{20}$$

whose last block is a linear function of the first two and thus changes ridge regularization as well as representation; the two-block control $x_{S'}^{\mathrm{2B}} = [x_{S'}; x_{S_0}]$ omits it. Single Ridge uses one predictor with the same data, features, and regularization and no disagreement penalty. It reaches 0.181 MAE and 54.08 F1, 0.38 below the full rule, and always predicting zero gain gives 0.205 MAE. The MLP control shares questions, features, labels, and development selection; its architecture is in the repository.

Acceptance search and outcomes. Development search on 100 questions per dataset (500 in total), disjoint from all training questions, covers $\kappa \in \{0, 0.25, 0.5, 0.75, 1, 1.5, 2, 3\}$ and $\tau \in \{0, 0.005, 0.01, 0.015, 0.02, 0.03, 0.04, 0.05, 0.075, 0.1\}$ on the $[-1, 1]$ gain scale

and selects $\kappa = 1$ and $\tau = 0.04$. Of the accepted swaps, 812 (full rule) and 777 (mean-only) leave F1 unchanged, with mean positive/negative changes of +0.115/−0.109 and +0.093/−0.135. For the full rule, $\kappa = 0$, and $\tau = 0$ with $\kappa = 1$, harmful/accepted counts are 127/1,427, 245/1,638, and 335/2,201 (8.90, 14.96, and 15.22%).

C Robustness, Transfer, and Cost Details

Support coverage and reader edits. Over BGE Top-2, DCR-RAG raises support recall from 90.90 to 94.30% on HotpotQA and from 81.20 to 86.55% on 2Wiki. Complete-support counts out of 1,000 for BGE Top-2, BGE+draft (independent/full), and the learned scorer (independent/full) are 823/850/861/874/892 on HotpotQA and 626/669/690/708/733 on 2Wiki; independent Top-2 with the learned scorer reaches 93.35 and 85.10% recall. On MuSiQue, BGE Top-2 and DCR-RAG fully cover 13 and 15 of 500 questions (2.60 and 3.00%, all two-hop); truncation retains all 15, and reader-visible recall is 48.15% versus 48.25% selected. Table 12 gives recall by decomposition position. On four-hop questions, fourth-position recall rises by 15.13 points [9.21, 21.05] and first-position recall falls by 7.89 [2.63, 13.82], while complete coverage does not change.

Reader edits fix the question, the other passage, and greedy decoding, and nonsupport donors match the token count of the full chat prompt. On MuSiQue, replacing a support or nonsupport passage ($n = 448/187$) changes F1 by −15.53 [−18.44, −12.67] and −1.70 [−4.01, 0.50], and deleting it ($n = 477/216$) by −13.34 [−16.01, −10.65] and −2.35 [−5.07, 0.49]; the paired comparison uses the 162 questions eligible for both. On HotpotQA and 2Wiki, support/nonsupport replacement changes F1 by −14.12/−1.96 and −16.56/−1.56.

Draft perturbations. Under strict draft EM, 653/347 drafts are correct/wrong on HotpotQA and 341/659 on 2Wiki. Support exclusion is measured against a reference pair, the independent Top-2 obtained with the question's natural draft: a question counts as an exclusion when a gold supporting passage in this pair is absent from the final selected pair. With natural drafts, exclusions can thus arise only from accepted replacements; with edited drafts, they can also arise from the changed ranking. Exclusion occurs for 41 of 994 natural-correct and 68 of 1,006 natural-wrong drafts (95% CIs [3.05, 5.55] and [5.37, 8.48]%), and a correct draft is followed by a wrong final answer in 49 of 994 cases (4.93%, [3.75, 6.46]). Entity and relation edits alter only the corresponding span of the same question's natural draft; exclusion then occurs in 81 of 826 (9.81%, [7.96, 12.02]) and 88 of 748 cases (11.76%, [9.65, 14.27]), with 97 and 104 final failures. Strict-EM failure rises from 5.21 and 5.08% to 11.74 and 13.90%, increases of 6.54 [4.11, 8.95] and 8.82 [5.99, 11.65] points.

Table 12: MuSiQue support recall (%) by decomposition position with Top-2 passages.

Type	n	Method	Hop 1	Hop 2	Hop 3	Hop 4
2-hop	24	BGE Top-2	87.50	54.17	–	–
		DCR-RAG	87.50	62.50	–	–
3-hop	324	BGE Top-2	76.23	29.63	41.98	–
		DCR-RAG	74.69	30.25	50.62	–
4-hop	152	BGE Top-2	71.71	13.82	23.03	23.68
		DCR-RAG	63.82	17.76	25.00	38.82

Table 13: Reader-transfer controls. Values in (a) are macro F1 (%), three-seed means ± sample SD.

(a) Macro F1 without and with the draft

Reader	No draft	With draft	ΔF_1	95% CI
Qwen2.5-1.5B	43.91 ± 0.08	45.35 ± 0.10	+1.44 ± 0.06	[0.61, 2.23]
Qwen2.5-3B	46.80 ± 0.13	48.60 ± 0.09	+1.80 ± 0.05	[0.91, 2.51]
Qwen2.5-7B	52.54 ± 0.06	54.47 ± 0.11	+1.93 ± 0.05	[1.12, 2.67]
Qwen2.5-14B	51.16 ± 0.09	53.14 ± 0.08	+1.98 ± 0.06	[1.14, 2.88]
Llama-3.1-8B	49.13 ± 0.14	50.87 ± 0.10	+1.74 ± 0.05	[1.01, 2.63]
Mistral-7B-v0.3	44.26 ± 0.11	45.60 ± 0.10	+1.34 ± 0.07	[0.46, 2.25]
InternLM3-8B	47.74 ± 0.09	49.62 ± 0.08	+1.88 ± 0.06	[0.96, 2.71]

(b) Outcomes of the 1,427 accepted swaps, seed 2094

Reader	Beneficial	No change	Harmful
Qwen2.5-1.5B	425	831	171
Qwen2.5-3B	448	816	163
Qwen2.5-7B	488	812	127
Qwen2.5-14B	476	812	139
Llama-3.1-8B	462	817	148
Mistral-7B-v0.3	402	843	182
InternLM3-8B	455	818	154

Table 14: Seed-wise macro F1 and effects (mean ± sample SD); the BGE independent route is fixed.

Route or effect	2094	2095	2096	Mean ± SD
BGE + draft, independent	49.11	49.11	49.11	49.11 ± 0.00
BGE + draft, full selector	50.00	49.92	50.18	50.03 ± 0.13
Learned scorer, independent	53.54	53.45	53.40	53.46 ± 0.07
Full DCR-RAG	54.46	54.37	54.59	54.47 ± 0.11
Mean-only (matched)	53.83	53.89	53.92	53.88 ± 0.05
Full minus mean-only	+0.63	+0.48	+0.67	+0.59 ± 0.10
Scorer effect, independent	+4.43	+4.34	+4.29	+4.35 ± 0.07
Scorer effect, full selector	+4.46	+4.45	+4.41	+4.44 ± 0.03
Selector effect, BGE	+0.89	+0.81	+1.07	+0.92 ± 0.13
Selector effect, learned scorer	+0.92	+0.92	+1.19	+1.01 ± 0.16
Interaction	+0.03	+0.11	+0.12	+0.09 ± 0.05

Transfer and cost protocols. CUDA-synchronized timing on two RTX 4090 GPUs (2Wiki on GPU1) includes inter-process communication and CPU loading. DCR-RAG takes 1.012/0.986/1.148 s (mean/median/95th percentile) and full-pool revision 0.976/0.949/1.105 s. On the timing subset, DCR-RAG improves on BGE by 5.15 F1 for 0.613 s. At $k = 1$, the selected MuSiQue passage is a support for 450 of 500 questions (22/292/136 at two/three/four hops); mean recall is $(22/2 + 292/3 + 136/4)/500 = 28.47\%$.

D Limitations

DCR-RAG relies on a draft whose errors can propagate into selection. Editing only the entity or relation of a draft raises strict-EM failures by 6.54 and 8.82 points (Section 6.1), and the scorer does not estimate draft reliability. The gain predictor needs gold answers and reader outputs for about 700 proposals per dataset, and leave-one-dataset-out fitting stays 0.28 F1 below target fitting. The fixed threshold ignores initial-pair quality, and the 4,072-token draft input may truncate low-ranked passages in larger pools. Finally, our pools come from benchmark paragraph sets and Wikipedia retrieval rather than live Web search.